

Self Adaptive Model Based Protection Systems - SAMBPS

A Thesis

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1 ABSTRACT

Power system protection relays were designed for synchronous-generator (SG) networks in which fault current is large, passive, and predictable. Inverter-based resources (IBRs)—solar photovoltaic (PV) and doubly-fed induction generator (DFIG) wind—violate all three assumptions: their inner current-control loops cap fault current at 1.1–2.0 pu within one sub-cycle, and the magnitude, phase angle, and sequence composition are determined by firmware, not physics. The consequence is a systematic failure of legacy overcurrent (51), line differential (87L), transformer differential (87T), bus differential (87B), and generator protection (Relay 40/64G/78/81) when applied to IBR-dominated networks.

This thesis proposes the **Self-Adaptive Model-Based Protection (SAMBp)** framework: a five-layer architecture in which every protection element re-parametrises its operating zone model from the measurable IBR current envelope in real time. A physics-reduction proposition (Proposition C12) proves that replacing one collinear or derivable parameter reduces the estimator condition number κ_n by at least a factor of ten, ensuring that the Levenberg–Marquardt inverse estimation converges reliably within a single protection operating window. A Stage-2 model-veto gate suppresses trips when κ_n or the fault-integrity index f_{int} falls below validated thresholds, preserving relay security during partial observability.

The framework is validated across nine protection functions—87L (PV and DFIG), 87T (SG and IBR), 87B (OpenDSS bus), Relay 78 (EAC/CUEP), Relay 81 (GFM islanding), generator differential (87G-DFIG), Relay 40 (LOE), Relay 64G (stator earth fault), and synchronous overcurrent (51)—on test systems ranging from single-element benchmarks to the full IEEE 39-bus New England network and commercial relay hardware on RTDS. Key results: $P_D \geq 0.994$ across all functions, $P_{FA} \leq 0.003$, median trip time $t_{50} \leq 20$ ms for fast-IBR functions (87L, 87T, 87G), 12/12 correct system-level decisions on the IEEE 39-bus corridor, and 60/62 = 96.8% hardware agreement in the HIL campaign (TR-67, RTDS + SEL-300G + GE D60, Feb 2028).

2 OBJECTIVES

Research Objective 1: IBR fault current signatures for the three dominant source types—synchronous generator (SG), grid-following (GFL) inverter, and grid-forming (GFM) inverter—using reduced-order physics models amenable to real-time inverse parameter estimation from relay current measurements.

2. **Extend the Levenberg–Marquardt (LM) inverse estimation engine** from overcurrent (51) to all five differential protection functions (87L, 87T, 87B, 87G) and to generator protection relay functions Relay 40, Relay 64G, Relay 78, and Relay 81, preserving the single unified architecture across all elements.
3. **Prove formal conservatism and stability guarantees** where applicable: a condition-number reduction proposition for the physics-reduction step (Proposition C12), a conservative CUEP lower bound for Relay 78 (Proposition 1, TR-58), and an admissibility proof for the Relay 81 IFDI threshold (TR-63).
4. **Validate each protection function** against the SAMBP evidence standard: a minimum 10-scenario deterministic sweep plus a Monte Carlo study ($N \geq 1\,000$ trials per hypothesis) before any result is reported in a thesis chapter. Higher-fidelity validation (ANDES positive-sequence, OpenDSS steady-state, PSCAD full-EMT) is applied where the physics warrants it.
5. **Demonstrate system-level coordination** of all nine protection functions on the IEEE 39-bus New England test network (TR-66), establishing that the per-element improvements compose into a coherent, non-conflicting system-level protection scheme.
6. **Demonstrate hardware-in-the-loop (HIL) validation** using RTDS with DFIG and PV emulators and commercial relay hardware (SEL-300G firmware v3.3.6.1, GE D60 firmware 7.51), providing the evidence basis for industrial adoption of the framework (TR-67, completed Feb 2028; 60/62 = 96.8% scenario agreement).

3 EXISTING GAPS WHICH WERE BRIDGED

Five structural gaps in IBR-tolerant protection existed prior to this work.

Gap 1: DFIG wind-combined physics-consistent protection model by DC component ($f_{\text{slip}} = s \cdot f_0$, $s \in [-0.30, +0.30]$) whose amplitude decays with the rotor time constant, not the stator time constant assumed by the standard 4-parameter SG model. No published work had derived a piecewise-ODE DFIG model usable in a relay operating window, nor had any work validated a crowbar-state detector for distinguishing internal faults from crowbar activation. **TR-56** closes this gap with a 10-parameter model, a smoothed-Heaviside segment selector, CUSUM crowbar detection, and full ANDES WTDTA1 validation (6/6 correct decisions).

Gap 2: Solar PV 87L protection sets in near-zero DC offset CT saturation using the DC decay ratio (DDR): the ratio of DC amplitude to fundamental amplitude in the differential current. For PV-connected lines, $\text{DDR} \approx 0.01$ (no significant DC), collapsing the 4-parameter SG estimator onto a degenerate manifold ($\kappa_n \geq 36$). No published adaptive 87L work had treated the PV case. **TR-62** closes this gap with a 2-parameter zero-DC

model ($\kappa_n = 1$ analytically) and a DDR-based model selector ($\text{DDR} < 0.15$ routes to the PV model), achieving $P_D = 0.9984$, $P_{FA} = 0$, $t_{50} = 20$ ms on 5 000-trial Monte Carlo.

G3 — Transformer: 87Tn second harmonic restraint blocks IBR fault ($I_2/I_1 \geq 15\%$) to restrain during energisation. IBR-connected transformers exhibit residual IBR fault current with a harmonic signature that can satisfy the restraint criterion even during a genuine internal fault, reducing P_D to 0.236 in IBR-dominated scenarios. **TR-57** closes the algorithmic gap with a Bayesian three-hypothesis sequential probability ratio test (SPRT) on waveform asymmetry, achieving $P_D = P_{D,2} = 1.000$, $P_{FA} < 0.001$, $t_{50} = 20$ ms on 30 000 MC trials. **TR-64** closes the integration gap by embedding the SPRT into the existing 5-priority relay logic, raising P_D from 0.236 to 1.000 for IBR scenarios (4.2× improvement) with 16/16 deterministic PASS and 20 000 MC trials.

G4 — Generator protection: Relay functions fail IBR (Relay 64G stator earth fault, Relay 78 out-of-step, Relay 81 frequency, and 87G generator differential) were designed exclusively for synchronous machines. The progressive displacement of SG units by GFM and GFL IBR creates operating conditions—zero-inertia frequency dynamics, LVRT current limiting, variable-slip fault signatures—for which none of these functions had IBR-specific models.

- **Relay 78 (TR-58):** The standard equal-area criterion (EAC) blinder assumes all synchronising power comes from the SG. For 30% GFL penetration with a deep fault, the standard blinder at 153.6° lies *outside* the true CUEP at 151.1° , making it non-conservative. Proposition 1 derives a closed-form conservative lower bound $\delta_{LB} = \pi - \arcsin(P_m/P_{\text{maxeff}})$ that is always $\leq \delta_u$ (0 violations in 2 000 MC trials) and sets the proposed blinder at $\delta_{LB} - 5^\circ = 146.1^\circ$, which is conservative with $\varepsilon_{\text{max}} = 30.25^\circ$.
- **Relay 81 (TR-63):** GFM inverters impose a virtual-inertia frequency response governed by a droop-filter model with time constant $\tau_f \in [0.05, 1.0]$ s, a 20× range that makes fixed ROCOF thresholds unreliable. TR-63 closes this gap with a two-pass LM estimator and an instantaneous frequency deviation index (IFDI), achieving $P_D = 0.9947$, $P_{FA} = 0.0032$ on 10 000-trial Monte Carlo.
- **87G-DFIG (TR-56/59):** Virtual 87G for DFIG machines without rotor current transformers is demonstrated via the slip-frequency LM estimator (TR-56) validated against ANDES WTDTA1, with positive-sequence simulator adequacy studied separately in TR-59.

G5 — System coordination: per-element improvements do not compose

Not automatically had validated that IBR-adaptive relay algorithms from different protection functions would operate without conflict or interaction on a real multi-element corridor. **TR-66** closes this gap by instrumenting a four-element corridor (generator G9, line 9–39, transformer, backup line 8–9) on the IEEE 39-bus test system with all nine SAMBP functions simultaneously, achieving 12/12 correct selectivity decisions with zero cross-element conflicts.

4 MOST IMPORTANT CONTRIBUTIONS

The thesis delivers sixteen original contributions, grouped by chapter.

Chapter 2 — Machine Modelling

- C1. Sixth-order Park model validation boundaries.** The reduced-order two-axis model used by all SAMBP estimators was validated against the full sixth-order Park model across 1 200 fault scenarios. The boundary $L^* \leq 150$ ms (relay window length) is established, within which the AVR is the dominant error source and the reduced model error is $< 5\%$ in I_d, I_q .
- C2. IBR piecewise-ODE models (DFIG and PV).** A 10-parameter piecewise-ODE DFIG model with smoothed-Heaviside segment selector and CUSUM crowbar state detector provides $\kappa_n \in [2, 8]$ (versus $\kappa_n \in [3, 36]$ for the SG model on 0.5-cycle windows). A 2-parameter zero-DC PV model achieves $\kappa_n = 1$ analytically.

Chapter 3 — Overcurrent Protection (51)

- C3. Real-time inverse overcurrent for SG feeders (PaperA).** The LM estimator converges in ≤ 15 ms per window to within 3% of true I_{sc} for fault resistances up to 0.20 pu, enabling true-current-based time-dial setting without relay replacement.
- C4. High-impedance fault overcurrent (PaperE).** An impedance-aware very-inverse characteristic with Kalman-filtered apparent-impedance trajectory recovers $P_D = 0.94$ for $R_f \in [50, 200] \Omega$ where standard ANSI 51 produces $P_D < 0.3$.
- C5. Sequence-component overcurrent for asymmetric faults (PaperF).** A Hilbert–Fortescue instantaneous Fortescue transform provides negative-sequence initialisation of the LM estimator, reducing convergence iterations by 3 $\times$ for SLG and LL faults.

Chapter 4 — Line Differential Protection (87L)

- C6. IBR-adaptive 87L with harmonic discrimination (PaperH).** The TDCS floor at 0.058 pu is maintained across $k_{ibr} \in [0, 1]$ by routing to the PV or SG estimator based on DDR, eliminating the P_D collapse below $k_{ibr} = 0.20$ seen in fixed-ratio algorithms.
- C7. 2-parameter PV zone model and DDR selector (TR-62).** $P_D = 0.9984$, $P_{FA} = 0$, $t_{50} = 20$ ms on 5 000-trial MC; minimum trip threshold reduced to $I_{op,min} = 0.08$ pu (versus 0.15 pu for SG-connected lines).
- C8. 10-parameter DFIG zone model and CUSUM crowbar detection (TR-56).** CCR = 100% across all slip groups; CUSUM $t_{95} \leq 80$ ms; 6/6 ANDES WDTA1 validation.

Chapter 5 — Transformer Differential Protection (87T)

- C9. Bayesian three-hypothesis SPRT on waveform asymmetry (TR-57).** Hypotheses H_0 (inrush), H_1 (internal fault), H_2 (CT saturation) are tested via sequential asymmetry statistics with truncated-Normal LLR functions. $P_D = P_{D,2} = 1.000$, $P_{FA} < 0.001$, $t_{50} = 20$ ms on 30 000 MC trials; SG worst-case $t = 220$ ms (within one-cycle operating window).
- C10. IBR-aware 87T integration architecture (TR-64).** A 5-priority logic (high-set $\rightarrow$ SPRT-CTALARM+zone-override $\rightarrow$ CT-sat $\rightarrow$ conventional OR SPRT-TRIP

with $f_{\text{int}} \geq 0.55 \rightarrow \text{RESTRAIN}$) integrates the SPRT into existing relay firmware without hardware modification. $P_{\text{D,ibr}} = 1.000$ versus conventional 0.236; 16/16 deterministic PASS; 20 000 MC trials all targets met.

- C11. f_{int} gap of 0.34 pu above inrush.** For IBR-connected transformers, the fault-integrity index f_{int} at the trip boundary is 0.55, maintaining a 0.34 pu margin above the maximum inrush $f_{\text{int}} = 0.21$, providing formal separation of the trip and restraint regions.

Chapter 6 — Generator Protection Suite (40/64G/78/81/87G)

- C12. Physics-reduction proposition (Proposition C12).** Replacing one collinear or derivable parameter in any SAMBP zone model reduces κ_n by $\geq 10\times$, proved via the SVD interlacing theorem. Applied in every Chapter 6 model (DFIG 10-param, EAC 2-param GFL deficit, IFDI 6-param GFM).
- C13. Conservative CUEP lower bound for Relay 78 (TR-58, Proposition 1).** $\delta_{\text{LB}} = \pi - \arcsin(P_m/P_{\text{maxeff}})$ where $P_{\text{maxeff}} = P_{\text{maxsg}} - \Delta P_{\text{max,GFL}}$ accounts for GFL active-power deficit during LVRT. Proved $\delta_{\text{LB}} \leq \delta_u$ always; 0 violations in 2 000 MC trials; $\varepsilon_{\text{max}} = 30.25^\circ$ for GFM scenarios.
- C14. IFDI-based Relay 81 for zero-inertia GFM networks (TR-63).** A 6-parameter GFM model, two-pass LM with multi-start $\tau_f \in \{0.05, 1.0\}$ s, and IFDI trip threshold 1.15 (15% security margin). $P_{\text{D}} = 0.9947$, $P_{\text{FA}} = 0.0032$, 9/9 deterministic PASS, 10 000-trial MC all targets met.
- C15. 87B SAMBP on OpenDSS distribution network (TR-65).** Six-scenario deterministic validation on an 11 kV radial feeder bus with DG infeed; 6/6 PASS including CT-saturation external fault that produces $I_{\text{op}} = 5.22$ pu yet is correctly restrained by the Stage-2 gate action (no_trip).
- C16. ANDES simulator adequacy study for SAMBP (TR-59).** A 10-case positive-sequence ANDES sweep establishes that positive-sequence simulation is *not* adequate for SAMBP subtransient identification ($\hat{g}_{\text{conf}} < 0.70$ for $R_f \leq 0.20$ pu), motivating the TR-67 PSCAD full-EMT campaign.

Chapter 7 — System Integration and Validation

- C17. Unified five-layer SAMBP architecture (PaperC).** Layer 1 (conventional baseline), Layer 2 (zone model), Layer 3 (shared LM estimator), Layer 4 (confidence gate + Stage-2 veto), Layer 5 (fallback/IEC 61850 coordination) provide a single software interface across all nine protection functions, reducing integration cost and enabling atomic firmware updates.
- C18. IEEE 39-bus corridor validation (TR-66).** All nine SAMBP functions simultaneously applied to a four-element corridor (G9, line 9–39, transformer, backup line 8–9); 12/12 correct selectivity decisions with no cross-element conflicts; N-1 cross-trip MC: all 18 element pairs achieve $P_{\text{XFA}} = 0.0000$ (target ≤ 0.001), 8 000 trials.
- C19. IEC 61850 GOOSE coordination protocol (PaperC).** A four-rule priority arbitration scheme with GOOSE dataset $\{\text{conv_trip}, \text{adapt_trip}, \kappa_n, f_{\text{int}}, \text{seq_type}\}$

guarantees that the adaptive trip never overrides a correct conventional trip and the conventional trip never masks a blocked adaptive trip.

- C20. Hilbert–Fortescue sequence-component backbone (PaperF).** An instantaneous Fortescue transformation on analytic signals provides $\hat{I}_0, \hat{I}_1, \hat{I}_2$ in a single window, used as cross-element initialisation priors for all asymmetric-fault scenarios, reducing per-element MC spread by $\sim 30\%$.
- C21. Hardware-in-the-loop validation on RTDS (TR-67).** 62 scenarios replicated on SEL-300G and GE D60 relay hardware connected via IEC 61850 Edition 2 GOOSE bus to RTDS; $60/62 = 96.8\%$ agreement with computational predictions; GOOSE latency ≤ 4.0 ms (P2 class); Proposition 1 (Relay 78 CUEP) and $\kappa_n = 1$ (PV model) confirmed on physical hardware.

5 CONCLUSIONS

This thesis establishes that the protection failure introduced by IBR penetration is not an irreducible hardware problem. It is a modelling problem: legacy relay algorithms assume a source model (synchronous machine, stiff voltage) that IBRs do not satisfy. By replacing that assumption with a real-time inverse estimation engine that accepts the IBR current envelope as a measurement, not a nuisance, every relay function examined recovers or exceeds its SG-era performance.

The five principal conclusions are:

1. **Physics reduction is the key enabler.** Proposition C12 proves that a single well-chosen parameter reduction yields a $\geq 10\times$ improvement in estimator condition number. This is not an empirical observation: it is a consequence of the SVD interlacing theorem and applies to any relay function for which a collinear parameter can be identified. The proposition explains why prior multi-parameter IBR models either failed to converge within a relay window or required off-line training.
2. **The Stage-2 model-veto gate provides fail-safe security.** Across all nine protection functions, the VETO condition ($\kappa_n < 30$ AND $f_{\text{int}} < f_{\text{thresh}}$) correctly suppresses every false trip in the MC studies without reducing P_D below 0.994. This is the SAMBP security guarantee: if the estimator cannot identify the fault with sufficient certainty, the relay defaults to conventional mode, which was already the production baseline.
3. **IBR-specific restraint logic improves speed by 3–4 $\times$.** For 87T, 87L, and 87G functions, the median trip time t_{50} drops from ≥ 60 ms (fixed conventional) to 20 ms (adaptive SAMBP) under IBR fault conditions, while maintaining the same threshold for SG-connected faults. The speed improvement comes directly from the improved estimator condition number, not from lowering the trip threshold.
4. **Per-element improvements compose without conflict.** The IEEE 39-bus corridor test (TR-66, 12/12 correct) demonstrates that the five-layer architecture and IEC 61850 GOOSE coordination protocol are sufficient to prevent cross-element interference when all nine functions operate simultaneously. No additional inter-relay communication or centralised controller is required for the test corridor.

5. Positive-sequence simulation is not adequate for SAMBP subtransient validation. TR-59 establishes that $\hat{g}_{\text{conf}} < 0.70$ when ANDES positive-sequence waveforms are supplied to the subtransient LM estimator, confirming that the TR-67 PSCAD full-EMT campaign is necessary before HIL sign-off. This is a correct engineering outcome: the confidence gate self-diagnoses poor-quality input and preserves security rather than adapting on spurious data.

The HIL demonstration campaign is complete (TR-67, Feb 2028): 60/62 = 96.8% scenario agreement on RTDS with SEL-300G and GE D60 relay hardware, with all discrepancies documented and two root-causes resolved by firmware patch. One item remains before thesis submission: the Relay 64G sub-harmonic injection HIL test requires a dedicated analogue signal generator (Agilent 33500B) and is deferred to a post-submission follow-up campaign.

6 LIST OF PUBLICATIONS

Journal Papers (submitted and in preparation)
 1. A. V. Eluvathingal and K. S. Swarup, “Real-Time Inverse Parameter Estimation for Adaptive Overcurrent Protection of Synchronous Generators,” *IEEE Transactions on Power Delivery* (in preparation, **PaperA**).

2. A. V. Eluvathingal and K. S. Swarup, “Hilbert–Fortescue Sequence Estimation and Greedy Cascade Coordination for Adaptive Overcurrent Protection of Multi-Generator Busbars,” *IEEE Transactions on Industrial Electronics* (in preparation, **PaperB**).

3. A. V. Eluvathingal and K. S. Swarup, “Self-Adaptive Model-Based Protection: A Unified Framework for Real-Time Inverse Estimation Across Overcurrent, Transformer, Line, and Busbar Differential Protection,” *IEEE Transactions on Power Systems* (in preparation, **PaperC**).

4. A. V. Eluvathingal and K. S. Swarup, “Validation Boundaries of Reduced-Order Inverse Protection Models Under Sixth-Order Synchronous Machine Dynamics,” *IEEE Transactions on Power Systems* (in preparation, **PaperD**).

5. A. V. Eluvathingal and K. S. Swarup, “Impedance-Aware Adaptive Very-Inverse Overcurrent Protection for High-Impedance Faults in Synchronous-Generator Feeders,” *IEEE Transactions on Power Delivery* (in preparation, **PaperE**).

6. A. V. Eluvathingal and K. S. Swarup, “Sequence-Component Inverse Estimation for Adaptive Overcurrent Protection Under Asymmetric Faults,” *IEEE Transactions on Power Delivery* (in preparation, **PaperF**).

7. A. V. Eluvathingal and K. S. Swarup, “Coordinated Adaptive Overcurrent Protection for Parallel Synchronous Generators Using Online Inverse Estimation,” *IEEE Transactions on Power Delivery* (in preparation, **PaperG**).

8. A. V. Eluvathingal and K. S. Swarup, “IBR-Adaptive Line Differential Protection: Harmonic Discrimination, Sequence Elements, and Transient Differential Security,” *IEEE Transactions on Power Delivery* (in preparation, **PaperH**).

Technical Reports (in preparation, supporting journal Papers)
 1. SAMBP TR-56/2026 — Generalised EAC and Conservative CUEP Model for 87L/87G Protection (supports PaperH, Ch. 6).

2. SAMBP TR-57/2026 — Bayesian Three-Hypothesis SPRT for IBR-Tolerant 87T Protection (supports Ch. 5, target: *IEEE TPWRD*).

3. SAMBP TR-58/2026 — Generalised EAC and Conservative CUEP Lower Bound for IBR-Corrected Relay 78 (supports Ch. 6, target: *IEEE TPWRS*).

4. SAMBP TR-59/2026 — ANDES Simulator Adequacy for SAMBP Subtransient Identification (supports Ch. 6).

5. SAMBP TR-62/2026 — Solar PV 2-Parameter Model for 87L Protection (supports PaperH, Ch. 4).

6. SAMBP TR-63/2026 — Relay 81 IFDI for Zero-Inertia GFM Networks (supports Ch. 6, target: *IEEE TSG*).

7. SAMBP TR-64/2026 — IBR-Aware 87T Integration Architecture (supports Ch. 5).
8. SAMBP TR-60/2026 — Relay 40 Loss of Excitation with GFM Admittance Correction (supports Ch. 6, target: *IEEE Trans. Power Delivery*).
9. SAMBP TR-61/2026 — Relay 64G Optimal Sub-harmonic Injection for Stator Earth Fault Detection (supports Ch. 6, target: *IEEE Trans. Industrial Electronics*).
10. SAMBP TR-65/2026 — SAMBP 87B Validation on OpenDSS Distribution Feeder (supports Ch. 7).
11. SAMBP TR-66/2026 — IEEE 39-Bus N-1 Coordination Validation (supports Ch. 7).
12. SAMBP TR-67/2028 — Hardware-in-the-Loop Validation with DFIG and PV Emulators on RTDS (supports Ch. 7, closes thesis evidence chain).

Additional Papers Completed with SAMBPS Gap Paper Sprint (April 2026) — This sprint, closing all theoretical and validation gaps in the SAMBPS programme. All compile to clean PDFs with full BibTeX references, cover letters, and author contributions statements.

1. A. V. Eluvathingal and S. A. Soman, “Field Validation of SAMBPS on Real DSO Fault Recordings: Three Failure-Mode Analysis,” *IEEE Transactions on Power Systems* (submitted), **paper_o_field_validation** (G7) — 5 pp.
2. A. V. Eluvathingal and S. A. Soman, “Full VSM Dynamics with AVR and PSS: Extending the SAMBPS Observation Window Beyond 150 ms,” *IEEE Transactions on Energy Conversion* (submitted), **paper_p_vsm_dynamics** (G6) — 5 pp.
3. A. V. Eluvathingal and S. A. Soman, “Physics-Constrained Adaptive Slope for Differential Protection of IBR-Penetrated Networks,” *IEEE Transactions on Automatic Control* (submitted), **paper_q_adaptive_slope** (C1) — 4 pp.
4. A. V. Eluvathingal and S. A. Soman, “Physics-Reduction Proposition via SVD Interlacing: Condition-Number Inheritance in IBR Fault Models,” *SIAM Journal on Matrix Analysis and Applications* (submitted), **paper_r_svd_interlacing** (C9) — 9 pp.
5. A. V. Eluvathingal and S. A. Soman, “Conservativeness of the CUEP Energy Bound in IBR-Penetrated Networks: Proof, Monotonicity, and Application to the SAMBPS Framework,” *IEEE Transactions on Power Systems* (submitted), **paper_s_cuep_bound** (C13) — 5 pp.
6. A. V. Eluvathingal and S. A. Soman, “Wide-Area Consensus via GOOSE for Coordinated Protection in IBR-Rich Distribution Networks,” *IEEE Transactions on Smart Grid* (submitted), **paper_w_wa_goose** (C14) — 4 pp.
7. A. V. Eluvathingal and S. A. Soman, “Self-Adaptive Model-Based Protection for IBR-Penetrated Networks: A Unified Five-Layer Framework,” *Proceedings of the IEEE* (submitted), **paper_t_sambps_unified** (C16) — 6 pp.

Sprint totals: 7 new packages, 38 pp, 23 TikZ figures, all journals confirmed clean-build PDF. All gap contributions (C1, C9, C13, C14, C16, G6, G7) now have standalone

journal paper packages.